

Law of Large Numbers (LLN)

- Time average $\rightarrow$ Statistical (expected) average
- Also called Empirical $\rightarrow$ True average

Setup

Let

$$\{X_i\}_{i \geq 1}$$

be a sequence of i.i.d. **random variables** with finite mean:

$$E[X_i] = \mu$$

Define:

$$S_n = \sum_{i=1}^n X_i$$

Statement (Weak LLN)

$$\frac{S_n}{n} \xrightarrow{P} \mu$$

Meaning:

Sample average $\rightarrow$ True mean (in probability)

Proof Sketch (Using Chebyshev)

Assume:

$$\text{Var}(X_i) = \sigma^2 < \infty$$

Then:

$$\text{Var}\left(\frac{S_n}{n}\right) = \frac{\sigma^2}{n}$$

Using Chebyshev inequality:

$$P\left(\left|\frac{S_n}{n} - \mu\right| \geq \epsilon\right) \leq \frac{\sigma^2}{n\epsilon^2}$$

As:

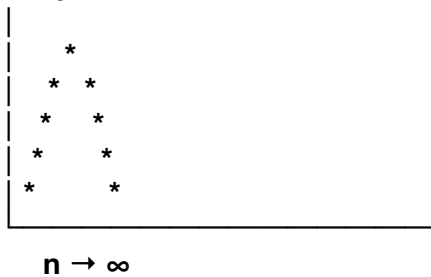
$$n \rightarrow \infty \Rightarrow \frac{\sigma^2}{n\epsilon^2} \rightarrow 0$$

So:

$$\frac{S_n}{n} \xrightarrow{P} \mu$$

Intuition Diagram

Sample Mean



→ converges to μ

Interpretation

- Average of observations stabilizes
- More data → better estimate of mean

Central Limit Theorem (CLT)

Let:

$$X_1, X_2, \dots, X_n$$

be i.i.d. with:

$$E[X_i] = \mu, \quad \text{Var}(X_i) = \sigma^2$$

Define:

$$S_n = \sum_{i=1}^n X_i$$

From LLN

$$\frac{S_n}{n} \xrightarrow{P} \mu$$

So:

$$\frac{S_n - n\mu}{n} \xrightarrow{P} 0$$

Question

How fast does:

$$S_n - n\mu$$

grow?

CLT STATEMENT

$$\frac{S_n - n\mu}{\sigma\sqrt{n}} \xrightarrow{D} N(0,1)$$

MEANING

Properly normalized sum $\rightarrow$ Normal distribution

Even if original distribution is NOT normal.

INSIGHT

- LLN $\rightarrow$ convergence of average
- CLT $\rightarrow$ distribution of fluctuations

VISUALIZATION

Original Distribution

skewed



Sample Mean Distribution

bell-shaped



Proof Idea (MGF Approach)

Let:

$$Z_n = \frac{S_n}{\sigma\sqrt{n}}$$

MGF of X_i :

$$M_X(t)$$

Then:

$$M_{Z_n}(t) = \left[M_X\left(\frac{t}{\sigma\sqrt{n}}\right) \right]^n$$

Taking log:

$$\log M_{Z_n}(t) = n \log M_X\left(\frac{t}{\sigma\sqrt{n}}\right)$$

Using Taylor expansion:

$$\log M_X(z) \approx \frac{\sigma^2 z^2}{2}$$

Substitute:

$$\log M_{Z_n}(t) \rightarrow \frac{t^2}{2}$$

Thus:

$$M_{Z_n}(t) \rightarrow e^{t^2/2}$$

Which is MGF of:

$$N(0,1)$$

Example: Bernoulli $\rightarrow$ Binomial

Let:

$$X_i \sim \text{Bern}(p)$$

Then:

$$\mu = p, \quad \sigma^2 = p(1 - p)$$

Sum:

$$S_n = \sum X_i \sim \text{Bin}(n, p)$$

CLT Result

$$\frac{S_n - np}{\sqrt{np(1 - p)}} \xrightarrow{D} N(0, 1)$$

Application Example

Coin Toss

- Toss fair coin 400 times
- Want:

$$P(S_{400} \leq 190)$$

$$\text{Exact Binomial} = P(S_{400} \leq 190) = \sum_{i=0}^{190} \binom{400}{i} (1/2)^{400} \approx 0.171$$

$$\text{CLT approximation} = \frac{S_{400} - 200}{\sqrt{100}} = \frac{190 - 200}{10} = -1$$

So:

$$P(S_{400} \leq 190) \approx P(Z \leq -1) = 0.1587$$

Insight - CLT gives very close approximation to binomial

NORMAL APPROXIMATION TO BINOMIAL

If: n large, p constant

$$\text{Bin}(n, p) \approx N(np, np(1 - p))$$

Poisson Approximation

If: $p \rightarrow 0, n \rightarrow \infty, np = \lambda$ (constant)

$$\text{Bin}(n, p) \rightarrow \text{Poisson}(\lambda)$$

Summary

LLN	Sample mean $\rightarrow$ true mean
CLT	Normal distribution emerges
Scaling	$\sqrt{n}$ is the correct scaling
Robustness	Works for any distribution (finite variance)